



G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, PUNE

AN EMPOWERED AUTONOMOUS INSTITUTE, AFFILIATED TO SAVITRIBAI PHULE PUNE UNIVERSITY,
PUNE

TAE 1 - REPORT

Diagnostic Laboratory Patient Records, Billing and Report Management Database System

Database Management System (DBMS)

Department of CSE (Data Science)

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G H RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, PUNE

Academic Year: 2026-2027

TAE-1 — E-Sports Tournament & Competitive Gaming Platform

Department of CSE – Data Science | Database Management Systems

Academic Year: 2026–27

1. Student Credentials

The project is prepared as an individual relational database design proposal for the E-Sports Tournament & Competitive Gaming Platform.

Field	Details
Student Name	Dhananjay Singh
Roll No.	P 13
Department	CSE – Data Science
Academic Year	2026–27
Course	Database Management Systems

2. Database System Title

System Title: E-Sports Tournament & Competitive Gaming Platform

The proposed system is a relational database designed to manage pro-gamers, VIP Battle Passes, game titles, esports leagues, esports teams, scheduled fixtures, tournament registrations, prize payouts, financial transactions, live match performances, and anti-cheat violation logs.

3. Problem Statement & System Requirements

3.1 Problem Statement

Competitive gaming platforms process millions of real-time events daily, including player profiles, squad rosters, tournament registrations, match fixtures, prize pool distributions, and automated anti-cheat telemetry. When these operations rely on fragmented legacy databases or unnormalized game server logs, serious data anomalies emerge: duplicate team rosters, inconsistent match scores, untraceable prize pool transactions, and delayed anti-cheat ban enforcement.

The E-Sports Tournament & Competitive Gaming Platform (ESTCGP) is proposed as a centralized, normalized relational database to unify all competitive gaming workflows. The system tracks players and VIP Battle Pass subscriptions, game titles and league structures, team rosters, match

fixtures, tournament entries, prize payouts, live match telemetry, and automated anti-cheat flag logs.

The database provides a single source of truth for fair tournament matchmaking, automated prize pool payouts, player performance ranking (MMR), and anti-cheat integrity audit trails.

3.2 Business Domain

The system operates within the competitive gaming and digital esports operations domain. The database models the interaction between players, gaming teams, esports leagues, match referees, automated server bots, anti-cheat detection software, and financial payment gateways.

The proposed system covers the player lifecycle from:

Player Lifecycle: *Player Registration & VIP Pass → Team Roster Join → Tournament Entry → Match Fixture Execution → Prize Payout & Performance Audit*

while maintaining operational tables required to monitor game server integrity and anti-cheat infractions.

3.3 Enterprise Entities

- **Player:** stores gamer identity, in-game handle (IGN), matchmaking ranking (MMR), and country.
- **VIP_Pass:** stores premium battle pass tier, reward perks, and expiration date (1:1 with Player).
- **Game_Title:** stores competitive video game catalog (e.g., Valorant, CS2), publisher, and rulesets.
- **Competitive_League:** stores tournament circuits, seasonal sponsors, and total prize pool allocations.
- **Team_Roster:** stores esports organization metadata, team tag, captain ID, and max squad capacity.
- **Match_Schedule:** stores scheduled match fixtures, assigned map/server, and scheduled start times.
- **Tournament_Entry:** records player/team tournament entries (resolves M:N between Player/Team and Match).
- **Prize_Payout:** stores official prize pool reward allocations based on final match placement.
- **Payout_Transaction:** records digital wallet/crypto/bank payouts and payment gateway hash references.
- **Match_Performance:** logs actual live match stats (KDA ratio, headshot %, combat score, winner tag).
- **AntiCheat_Violation:** logs anti-cheat engine flags (aimbot, wallhack), severity score, and ban status.

3.4 Business Rules & Constraints

- Every player must possess a unique Player_ID, unique Email, and unique In-Game Name (IGN).
- A player can hold at most one active VIP Battle Pass record (1:1 relationship).
- A competitive league belongs to exactly one game title, but a game title can host many leagues.
- A team roster belongs to one esports organization and must have a designated captain.
- A scheduled match fixture must assign two distinct teams (Team_A <> Team_B).
- A player can enter multiple tournament matches; each entry links one player/team to one scheduled match.
- Every prize payout generates exactly one financial payout transaction record (1:1 relationship).
- A live match performance log can generate zero or more anti-cheat violation flags.
- Primary keys must be unique and non-null; foreign keys must reference existing parent records.
- Monetary prize amounts, MMR ratings, and KDA stats must not contain negative values.

3.5 Primary Operational Transactions

1. Player & VIP Pass Management

- Register gamer profiles, update MMR ranks, and issue VIP battle passes.

2. League & Team Roster Management

- Register esports leagues, create team rosters, and assign team captains.

3. Tournament Entry & Matchmaking

- Schedule match fixtures, verify team eligibility, and record tournament entries.

4. Prize Payout & Payment Settlement

- Calculate prize pool distributions, generate payout receipts, and log transaction reference hashes.

5. Match Telemetry & Anti-Cheat Audit

- Log real-time match stats (KDA, combat score) and auto-flag anti-cheat violations.

6. Analytical Reporting

- Rank top players by MMR, calculate win/loss ratios per team, and analyze anti-cheat detection rates.

3.6 System Requirements

Functional Requirements

- Register and maintain gamer profiles, team rosters, and VIP battle passes.
- Manage game titles, league structures, and scheduled match fixtures.
- Record match entries, log live telemetry stats, and execute prize payouts.
- Flag anti-cheat violations and audit banned player accounts.

Data and Integrity Requirements

- Primary keys shall uniquely identify records and shall not be NULL.
- Foreign keys shall maintain valid relationships between parent and child relations.
- Domain constraints shall restrict game modes, payment methods, and KDA stats to valid values.
- The relational design shall reduce redundancy through normalization up to 3NF/BCNF.

4. Literature & System Reference

4.1 Literature Review

The proposed E-Sports Tournament & Competitive Gaming Platform is based on concepts from relational database design, online matchmaking architectures, and anti-cheat telemetry tracking.

Title: Reference 1 — Esports Analytics and Competitive Gaming Information Systems

Research on competitive gaming architectures highlights the necessity of centralizing player profiles, match histories, and team rosters into structured relational databases to support real-time skill rating (MMR) calculations and fair matchmaking.

Relevance to the proposed system: This supports the integration of Player, Team_Roster, Competitive_League, and Match_Schedule entities in a unified relational schema.

Title: Reference 2 — Automated Tournament Management and Digital Payout Gateways

Studies on online esports platforms demonstrate the importance of atomic transactional workflows for tournament registrations, entry fee processing, and automated prize pool distribution to prevent fraud and payment discrepancies.

Relevance to the proposed system: This provides the foundation for the Tournament_Entry → Prize_Payout → Payout_Transaction workflow used in ESTCGP.

Title: Reference 3 — Anti-Cheat Telemetry and Database Integrity Verification

Database literature on game security emphasizes logging live telemetry metrics alongside anti-cheat flags to establish tamper-proof audit trails for competitive integrity enforcement.

Relevance to the proposed system: These principles guide the integration of Match_Performance and AntiCheat_Violation tables in 3NF/BCNF.

4.2 System Reference

Existing esports platforms (such as FACEIT, ESL Play, and Battlefy) integrate operational areas including:

- Player identity and MMR ranking
- Team roster management
- League and tournament scheduling
- Automated bracket and match execution
- Prize pool distribution and payment logs
- Anti-cheat telemetry and ban logging

The proposed ESTCGP adopts these operational concepts to demonstrate ER/EER modelling, relational schema conversion, normalization, keys, constraints, and database design principles.

5. Conceptual Schema Design — ER/EER Diagram

5.1 Enterprise Entities and Main Attributes

- **Player:** Player_ID (PK), First_Name, Last_Name, Gamer_Handle_IGN, Gender, DOB, Email, Phone, Country, MMR_Rating
- **VIP_Pass:** Pass_ID (PK), Player_ID (FK), Pass_Tier, Start_Date, Expiry_Date, Pass_Status, Reward_Points
- **Game_Title:** Game_ID (PK), Game_Title, Publisher, Genre, Max_Team_Size, Ruleset_Ver
- **Competitive_League:** League_ID (PK), Game_ID (FK), League_Name, Season_Year, Total_Prize_Pool, Sponsor_Name
- **Team_Roster:** Team_ID (PK), Team_Name, Team_Tag, Captain_Player_ID (FK), Region, Created_Date
- **Match_Schedule:** Match_ID (PK), League_ID (FK), Team_A_ID (FK), Team_B_ID (FK), Server_Region, Scheduled_Time, Match_Status
- **Tournament_Entry:** Entry_ID (PK), Player_ID (FK), Team_ID (FK), Match_ID (FK), Entry_Date, Seed_Rank, Entry_Status | Composite Candidate Key: Player_ID + Match_ID + Entry_Date
- **Prize_Payout:** Payout_ID (PK), Entry_ID (FK), Placement_Rank, Prize_Amount, Tax_Deducted
- **Payout_Transaction:** Txn_ID (PK), Payout_ID (FK), Payment_Date, Amount_Paid, Payment_Method, Txn_Ref_Hash, Txn_Status
- **Match_Performance:** Perf_ID (PK), Match_ID (FK), Player_ID (FK), KDA_Ratio, Combat_Score, Headshot_Pct, Outcome
- **AntiCheat_Violation:** Violation_ID (PK), Perf_ID (FK), Cheat_Type, Flag_Confidence_Pct, Action_Taken, Logged_Time

5.2 Relationships & Cardinalities

Relationship	Cardinality
Player — VIP_Pass	1:1
Game_Title — Competitive_League	1:N
Competitive_League — Match_Schedule	1:N
Team_Roster — Match_Schedule (Team A / Team B)	1:N
Player — Tournament_Entry	1:N
Team_Roster — Tournament_Entry	1:N
Match_Schedule — Tournament_Entry	1:N
Tournament_Entry — Prize_Payout	1:1
Prize_Payout — Payout_Transaction	1:1
Match_Schedule — Match_Performance	1:N
Player — Match_Performance	1:N
Match_Performance — AntiCheat_Violation	1:N
Player — Match_Schedule	M:N, resolved through Tournament_Entry

The M:N relationship between Player and Match_Schedule is resolved through the associative entity Tournament_Entry:

Resolution: *Player 1:N Tournament_Entry N:1 Match_Schedule*

5.3 EER Diagram

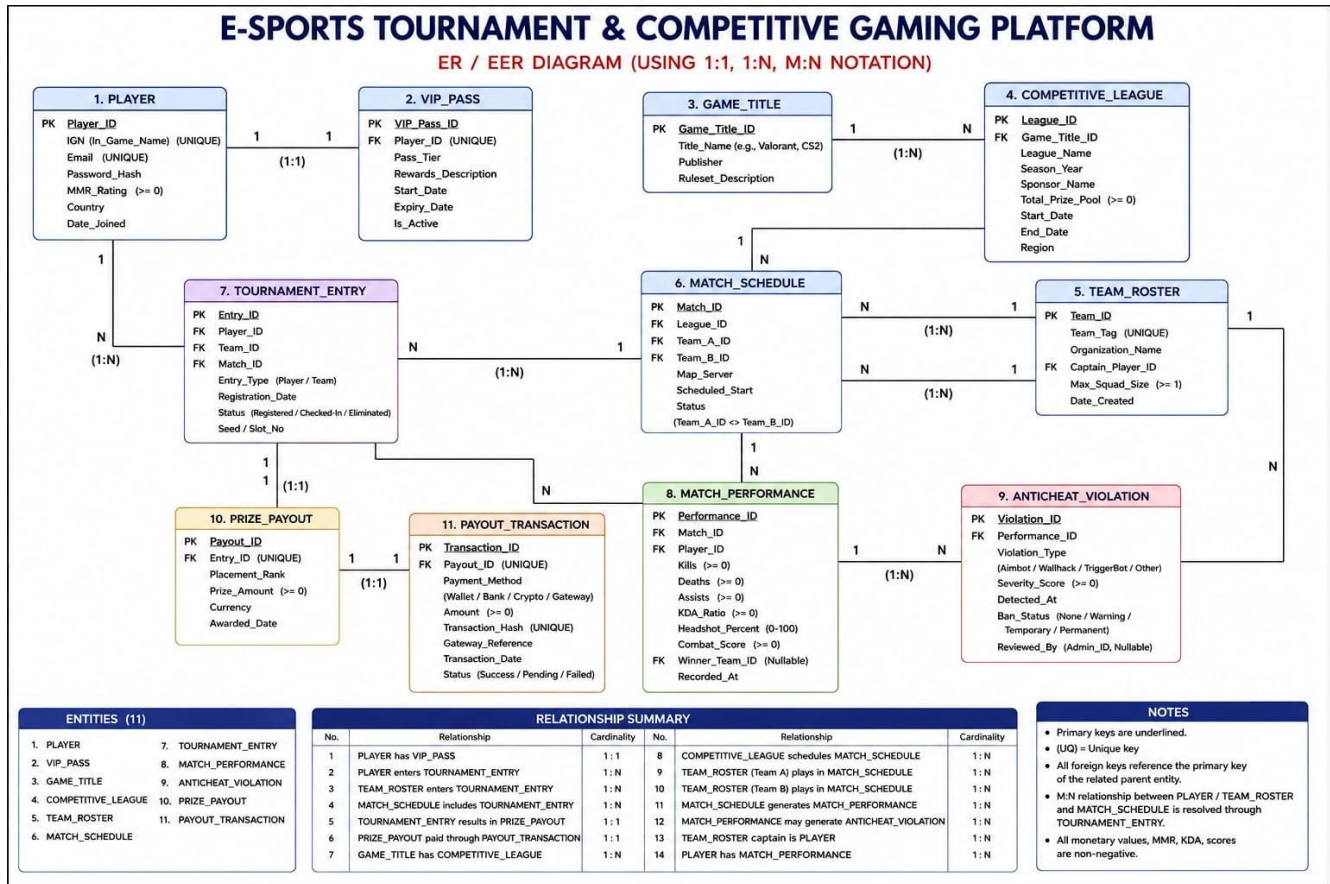


Figure 5.1: EER Diagram of E-Sports Tournament & Competitive Gaming Platform

- All 11 entities
- Entity attributes
- Primary Keys
- Foreign Keys
- Composite candidate key
- Named relationships
- 1:1, 1:N and M:N cardinalities
- M:N resolution through Tournament_Entry
- Relationship summary/legend

6. Relational Schema & Normalisation Plan

6.1 Relational Schema — Attributes, Data Types & Keys

6.1.1 PLAYER

Attribute	Data Type	Key / Constraint
Player_ID	VARCHAR(10)	PK, NOT NULL
First_Name	VARCHAR(50)	NOT NULL
Last_Name	VARCHAR(50)	NOT NULL
Gamer_Handle_IGN	VARCHAR(30)	UNIQUE, NOT NULL
Gender	VARCHAR(10)	CHECK
DOB	DATE	—
Email	VARCHAR(100)	UNIQUE, NOT NULL
Phone	VARCHAR(15)	—
Country	VARCHAR(50)	NOT NULL
MMR_Rating	INTEGER	CHECK (MMR_Rating >= 0)

6.1.2 VIP_PASS

Attribute	Data Type	Key / Constraint
Pass_ID	VARCHAR(10)	PK, NOT NULL
Player_ID	VARCHAR(10)	FK → PLAYER, UNIQUE, NOT NULL
Pass_Tier	VARCHAR(20)	CHECK
Start_Date	DATE	NOT NULL
Expiry_Date	DATE	NOT NULL
Pass_Status	VARCHAR(20)	CHECK
Reward_Points	INTEGER	CHECK (Reward_Points >= 0)

6.1.3 GAME_TITLE

Attribute	Data Type	Key / Constraint
Game_ID	VARCHAR(10)	PK, NOT NULL
Game_Title	VARCHAR(100)	UNIQUE, NOT NULL
Publisher	VARCHAR(100)	NOT NULL
Genre	VARCHAR(30)	NOT NULL
Max_Team_Size	INTEGER	CHECK (Max_Team_Size > 0)
Ruleset_Ver	VARCHAR(15)	NOT NULL

6.1.4 COMPETITIVE_LEAGUE

Attribute	Data Type	Key / Constraint
League_ID	VARCHAR(10)	PK, NOT NULL
Game_ID	VARCHAR(10)	FK → GAME_TITLE, NOT NULL
League_Name	VARCHAR(150)	NOT NULL
Season_Year	INTEGER	CHECK (Season_Year >= 2020)
Total_Prize_Pool	NUMERIC(12,2)	CHECK (Total_Prize_Pool >= 0)
Sponsor_Name	VARCHAR(100)	—

6.1.5 TEAM_ROSTER

Attribute	Data Type	Key / Constraint
Team_ID	VARCHAR(10)	PK, NOT NULL
Team_Name	VARCHAR(100)	UNIQUE, NOT NULL
Team_Tag	VARCHAR(10)	UNIQUE, NOT NULL
Captain_Player_ID	VARCHAR(10)	FK → PLAYER, NOT NULL
Region	VARCHAR(30)	NOT NULL

Created_Date	DATE	NOT NULL
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6.1.6 MATCH_SCHEDULE

Constraint: Team_A_ID and Team_B_ID must be different (Team_A <> Team_B).

Attribute	Data Type	Key / Constraint
Match_ID	VARCHAR(10)	PK, NOT NULL
League_ID	VARCHAR(10)	FK → COMPETITIVE_LEAGUE, NOT NULL
Team_A_ID	VARCHAR(10)	FK → TEAM_ROSTER, NOT NULL
Team_B_ID	VARCHAR(10)	FK → TEAM_ROSTER, NOT NULL
Server_Region	VARCHAR(30)	NOT NULL
Scheduled_Time	TIMESTAMP	NOT NULL
Match_Status	VARCHAR(20)	CHECK

6.1.7 TOURNAMENT_ENTRY

Composite Candidate Key: (Player_ID, Match_ID, Entry_Date)

Attribute	Data Type	Key / Constraint
Entry_ID	VARCHAR(10)	PK, NOT NULL
Player_ID	VARCHAR(10)	FK → PLAYER, NOT NULL
Team_ID	VARCHAR(10)	FK → TEAM_ROSTER, NOT NULL
Match_ID	VARCHAR(10)	FK → MATCH_SCHEDULE, NOT NULL
Entry_Date	DATE	NOT NULL
Seed_Rank	INTEGER	CHECK (Seed_Rank > 0)
Entry_Status	VARCHAR(20)	CHECK

6.1.8 PRIZE_PAYOUT

Attribute	Data Type	Key / Constraint
Payout_ID	VARCHAR(10)	PK, NOT NULL
Entry_ID	VARCHAR(10)	FK → TOURNAMENT_ENTRY, UNIQUE, NOT NULL
Placement_Rank	INTEGER	CHECK (Placement_Rank > 0)
Prize_Amount	NUMERIC(10,2)	CHECK (Prize_Amount >= 0)
Tax_Deducted	NUMERIC(10,2)	CHECK (Tax_Deducted >= 0)

6.1.9 PAYOUT_TRANSACTION

Attribute	Data Type	Key / Constraint
Txn_ID	VARCHAR(10)	PK, NOT NULL
Payout_ID	VARCHAR(10)	FK → PRIZE_PAYOUT, UNIQUE, NOT NULL
Payment_Date	DATE	NOT NULL
Amount_Paid	NUMERIC(10,2)	CHECK (Amount_Paid >= 0)
Payment_Method	VARCHAR(30)	CHECK
Txn_Ref_Hash	VARCHAR(64)	UNIQUE, NOT NULL
Txn_Status	VARCHAR(20)	CHECK

6.1.10 MATCH_PERFORMANCE

Attribute	Data Type	Key / Constraint
Perf_ID	VARCHAR(10)	PK, NOT NULL
Match_ID	VARCHAR(10)	FK → MATCH_SCHEDULE, NOT NULL
Player_ID	VARCHAR(10)	FK → PLAYER, NOT NULL
KDA_Ratio	NUMERIC(5,2)	CHECK (KDA_Ratio >= 0)

Combat_Score	INTEGER	CHECK (Combat_Score >= 0)
Headshot_Pct	NUMERIC(5,2)	CHECK (Headshot_Pct BETWEEN 0 AND 100)
Outcome	VARCHAR(10)	CHECK

6.1.11 ANTICHEAT_VIOLATION

Attribute	Data Type	Key / Constraint
Violation_ID	VARCHAR(10)	PK, NOT NULL
Perf_ID	VARCHAR(10)	FK → MATCH_PERFORMANCE, NOT NULL
Cheat_Type	VARCHAR(50)	NOT NULL
Flag_Confidence_Pct	NUMERIC(5,2)	CHECK (Flag_Confidence_Pct BETWEEN 0 AND 100)
Action_Taken	VARCHAR(30)	CHECK
Logged_Time	TIMESTAMP	NOT NULL

6.1.12 Summary of Keys

Table	Primary Key	Foreign Keys
PLAYER	Player_ID	—
VIP_PASS	Pass_ID	Player_ID
GAME_TITLE	Game_ID	—
COMPETITIVE_LEAGUE	League_ID	Game_ID
TEAM_ROSTER	Team_ID	Captain_Player_ID
MATCH_SCHEDULE	Match_ID	League_ID, Team_A_ID, Team_B_ID
TOURNAMENT_ENTRY	Entry_ID	Player_ID, Team_ID, Match_ID
PRIZE_PAYOUT	Payout_ID	Entry_ID
PAYOUT_TRANSACTION	Txn_ID	Payout_ID

MATCH_PERFORMANCE	Perf_ID	Match_ID, Player_ID
ANTICHEAT_VIOLATION	Violation_ID	Perf_ID

6.1.13 Integrity Constraints

- Primary keys must be unique and NOT NULL.
- Foreign keys must reference valid parent records.
- In-Game Handle (IGN), player email, team name, team tag, and transaction hash must be unique.
- Prize amounts, MMR rating, KDA ratio, and combat score cannot be negative.
- Team A and Team B in a match schedule must be different (Team_A <> Team_B).
- Foreign-key relationships prevent orphan records.
- Entry_ID is unique in PRIZE_PAYOUT and Payout_ID is unique in PAYOUT_TRANSACTION to enforce 1:1 relationships.

6.2 Unnormalised Form (UNF)

The original 1,250-record raw dataset contains player, VIP pass, game title, league, team, match, entry, payout, transaction, performance, and anti-cheat information in a single wide operational structure.

[illegible]

Link of raw dataset :
<https://docs.google.com/spreadsheets/d/1xhmsqsBo0uhB6Cgmd8q3Sgcw6u1paSyr/edit?usp=sharing&ouid=111286422379396777335&rtpof=true&sd=true>

ESPORTS_PLATFORM_RAW(Player_ID, First_Name, Last_Name, Gamer_Handle_IGN, Gender, DOB, Email, Phone, Country, MMR_Rating, Pass_ID, Pass_Tier, Pass_Status, Game_ID, Game_Title, Publisher, League_ID, League_Name, Total_Prize_Pool, Team_ID, Team_Name, Team_Tag, Match_ID, Server_Region, Scheduled_Time, Entry_ID, Entry_Date, Seed_Rank, Payout_ID, Placement_Rank, Prize_Amount, Txn_ID, Payment_Method, Txn_Ref_Hash, Perf_ID, KDA_Ratio, Combat_Score, Headshot_Pct, Violation_ID, Cheat_Type, Flag_Confidence_Pct)

Problems in UNF

- Repeated player demographic information
- Repeated game title and league metadata
- Repeated team roster details
- Mixed match fixture, payout, and telemetry logs
- Multi-valued perk fields
- Missing values and uncleaned gamer handles
- Inconsistent country representations
- Update, insertion, and deletion anomalies

6.3 First Normal Form (1NF)

The raw structure is transformed into 1NF by ensuring:

- Each attribute contains an atomic value.
- Repeating groups are removed.
- Each record is uniquely identifiable.
- Multi-valued fields are separated into atomic values.
- Missing and inconsistent representations are standardized.

Atomic Conversion Atomic Conversion Example:

UNF: Gamer_Handle_IGN = "ShadowNinja [EU_Pro]" is split into clean Gamer_Handle_IGN and Region.

Composite primary key established: {Entry_ID, Player_ID, Match_ID}.

6.4 Second Normal Form (2NF)

The relation is already in 1NF, but attributes depend on different identifiers.

Dependencies Identified: Examples of Partial Dependencies:

- Player_ID → First_Name, Last_Name, Gamer_Handle_IGN, Email, Country, MMR_Rating
- Game_ID → Game_Title, Publisher, Genre, Max_Team_Size
- Team_ID → Team_Name, Team_Tag, Captain_Player_ID, Region
- Match_ID → League_ID, Team_A_ID, Team_B_ID, Scheduled_Time
- Entry_ID → Player_ID, Team_ID, Match_ID, Entry_Date, Seed_Rank

Entity-specific details are separated into 11 dedicated relations.

6.5 Third Normal Form (3NF)

Transitive dependencies are removed.

For example: Match_ID → League_ID and League_ID → Game_ID, League_Name. Therefore, league details are stored in COMPETITIVE_LEAGUE, while MATCH_SCHEDULE contains only League_ID.

Similarly:

- VIP pass details are separated into VIP_PASS.
- Game title details are separated into GAME_TITLE.
- Payout transaction details are separated into PAYOUT_TRANSACTION.
- Anti-cheat violation logs are separated into ANTICHEAT_VIOLATION.

This eliminates transitive redundancy.

6.6 BCNF Verification

The principal functional dependencies are:

Relation	Functional Dependency
PLAYER	Player_ID → player attributes
VIP_PASS	Pass_ID → pass attributes
GAME_TITLE	Game_ID → game attributes
COMPETITIVE_LEAGUE	League_ID → league attributes
TEAM_ROSTER	Team_ID → team attributes
MATCH_SCHEDULE	Match_ID → match attributes
TOURNAMENT_ENTRY	Entry_ID → entry attributes
PRIZE_PAYOUT	Payout_ID → payout attributes
PAYOUT_TRANSACTION	Txn_ID → transaction attributes
MATCH_PERFORMANCE	Perf_ID → performance attributes
ANTICHEAT_VIOLATION	Violation_ID → violation attributes

The determinants are candidate keys in their respective relations. Therefore, the final schema satisfies BCNF.

6.7 Normalisation Summary

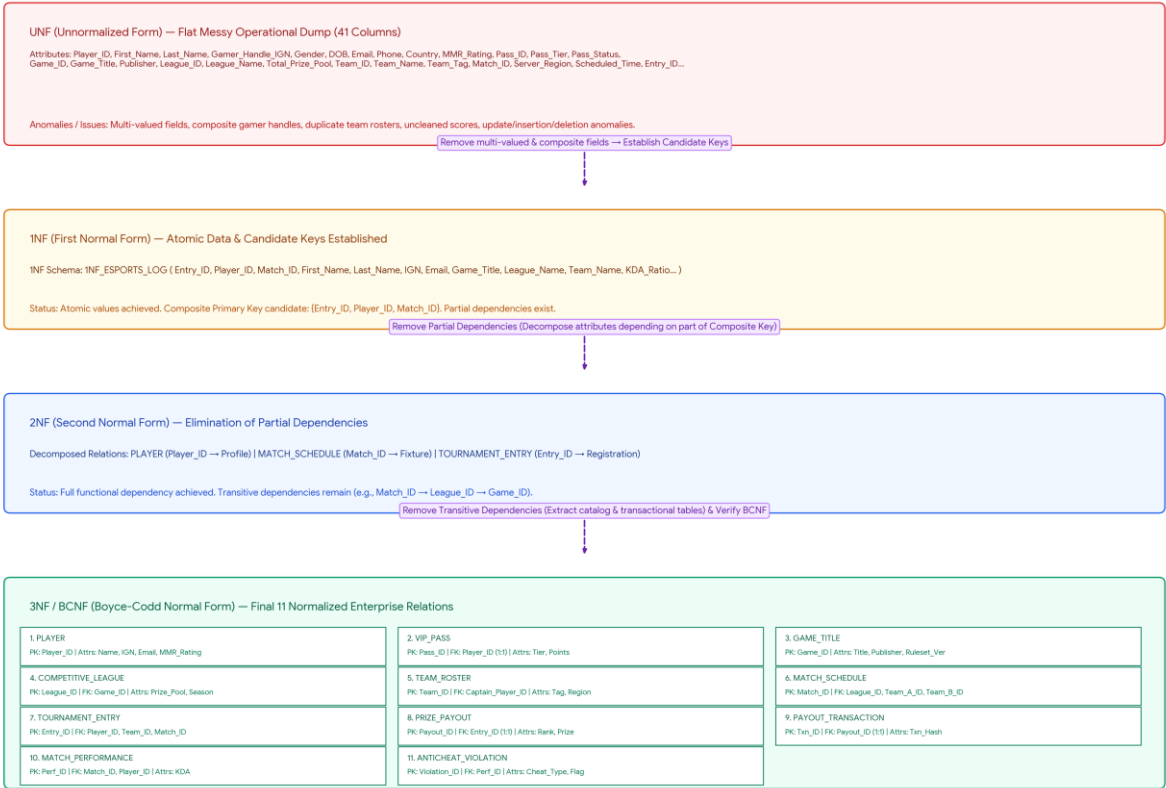
Stage	Action	Result
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UNF	Raw operational dataset containing repeating groups and redundant player, game, team, and match data across 41 attributes.	Unnormalized flat log structure with multi-valued fields and update/deletion anomalies.
1NF	Remove repeating groups, separate multi-valued perk fields into atomic values, and clean gamer handle representations.	Atomic values achieved; composite primary key candidate established: {Entry_ID, Player_ID, Match_ID}.
2NF	Remove partial dependencies on composite candidate key by decomposing non-key attributes into separate functional relations.	Player, Game, Team, and Match operational details decomposed into dedicated relations.
3NF	Remove transitive dependencies (e.g., Match_ID -> League_ID -> Game_ID; Entry_ID -> Payout_ID -> Txn_ID).	Catalog tables (GAME_TITLE, LEAGUE, TEAM) and transactional tables (PAYOUT, TRANSACTION, VIOLATION) separated.
BCNF	Verify that every determinant in all non-trivial functional dependencies is a candidate superkey.	Final 11 relations satisfy BCNF with full entity, referential, and domain integrity enforced.

Final Normalized Relations

Relations: PLAYER | VIP_PASS | GAME_TITLE | COMPETITIVE_LEAGUE | TEAM_ROSTER | MATCH_SCHEDULE | TOURNAMENT_ENTRY | PRIZE_PAYOUT | PAYOUT_TRANSACTION | MATCH_PERFORMANCE | ANTICHEAT_VIOLATION

FIGURE 6.2: NORMALIZATION PROCESS & RELATIONAL ARCHITECTURE
E-Sports Tournament & Competitive Gaming Platform (UNF to BCNF Transformation)



7. Target RDBMS Software

7.1 Selected RDBMS

Target RDBMS: PostgreSQL

PostgreSQL is proposed as the target relational database management system for Phase II (TAE-2) implementation of the E-Sports Tournament & Competitive Gaming Platform.

The relational schema, business rules, keys, relationships, integrity constraints, and normalization strategy developed in TAE-1 will be carried forward to the SQL implementation in TAE-2 without changing the approved database domain or design.